

The Basic idea of the paper, in a flow

The initial problem is optimal sensor placement for detecting stochastic target trajectories in Barrier coverage system. The first paper, previously, considered targets as stochastic points crossing a 1D barrier. To, make it a bit more practical, the discussed paper considers movement in a 2D space, where it approximates random ship trajectories (in reality ships/targets don't travel in straight lines and do take random turns and shifts) into straight lines (Under the assumption that the region of interest is small, the ship trajectories can be approximated as straight line segments.). The lines were modeled as a poisson line process.

After that each trajectory line is represented by a unique point using a bijective transformation in a representation space. After these points are achieved in a representation space, they become easy to process mathematically, and to estimate intensity, since stochastic target routes are tricky to deal with directly. However, these points have another complexity and that is they are not uniform in the discussed space, while some trajectory types are common and some are very rare within the representation space. Here comes the use of LGCLP, that estimates the traffic trajectory intensity in the representation space, showing which traffic trajectory types are frequent and which are rare. The estimated trajectory intensity is then used to compute the void probability, which measures the likelihood that a target trajectory is not detected by the sensor network. It then uses greedy sensor placement to minimize missed trajectories.

The mathematical approaches and problem facing of the discussed paper

The previous part was a story flowchart of how the paper solves the problem. This part discusses the mathematical formulation and technical approaches. The first paper (1D) tries to place sensors optimally using a nonhomogeneous point process and LGCP estimated intensity from historical data. The limitation was that targets were represented only as points crossing the barrier, so the actual trajectory direction and movement path were not captured.

Now the discussed paper, after assuming approximation lines from curved/random trajectories in an original space, represents the collection of trajectory lines following a Poisson Line Process and then mapped into a representation space. In this representation space every line is transformed to a unique point with bijective transformation with an angle and distance value. So, with these many points the next issue becomes they are random and not uniformly spread, with some common and rare trajectory types. So, next approach is trajectory intensity estimation with LGCLP. To efficiently estimate the intensity function from the historical data, the paper utilizes INLA to estimate LGCLP intensity efficiently.

After having trajectory intensity, a sensor detection model is introduced, a function that uses the target location and the sensor's position. The distance variety decides the probability of detection. Using this detection function models the probability that one sensor fails to detect one trajectory and later extends to the whole network of sensors to derive the probability that all sensors fail to detect one trajectory.

This leads to the point where void probability approximation comes in, which measures the likelihood of missed targets. Sensor locations are determined by minimizing this approximation by an iterative greedy search to ensure a better detection performance.

Possible Extension Idea

The discussed paper proposes sensor placement based on trajectories from historical data. In reality, ships might not follow as frequently in the historical routes due to various external factors like weather/traffic conditions, route/operation/port changes. To address these an extension is proposed with a prediction and uncertainty aware sensor network placement. Instead of relying only on historical data, future trajectory patterns are predicted and combined with an uncertainty model. Sensor placement is then optimized with the additional uncertainty consideration. Although the discussed paper handles uncertainty to estimate historical traffic intensity, this extension idea considers future traffic deviation before sensor placement.

Now this direction can lead to a possible future research direction with more condition/context aware uncertainty and sensor placement against that. Separate trajectory intensity models for different factors like weather, traffic, port operation, season change etc. can create a robust, need-based spontaneous model. AIS, ship, satellite companies or any other relevant sources can help provide the data pattern.

Technical approach story

The proposed extension keeps the main approach of the discussed paper. The same ideas and approaches like, line representation> bijective transformation> representation space> sensor detection model> void probability idea> greedy search are used as the base workflow.

The historical data were generated as random straight lines with small variations in slopes and intercepts. Then the same base workflow was followed.

For the prediction part, a future route prediction step is introduced. In real problem solving models like ARIMA time series, Kalman filter, LSTM deep learning models can be used for this kind of prediction. But for the toy simulation, first we tried a linear regression but the result lines and points were too narrow and not spread for lines or cloud-like after bijective transformation. Then changed the slope and the intercept value of each line to mimic a toy prediction. The purpose behind this step is to not measure the future routes accurately, but to test the changed sensor location under shifted future route pattern.

For the Uncertainty story- trajectory uncertainty might occur due to many external factors mentioned previously. Since there is no real AIS data available a simplified uncertainty model was used for toy simulation.

For each ship it's very tricky to find traffic deviation causes, and patterns, be the reason weather, port or operation change. The uncertainty awareness presents the scope to say that route parameters may vary within a plausible range. Historical deviation pattern and prediction error can contribute as inputs for uncertainty awareness models. With AIS data Gaussian Process based uncertainty prediction can be used. For the toy simulation just like before, uncertainty is introduced with slope and intercepts of the relevant lines.

With historical and predicted trajectories, several combinations of uncertainty can be observed. For example, uncertainty with historical trajectories, uncertainty with predicted trajectories and

finally “historical+predicted” trajectories. The “historical+predicted” routes are combined as a route set and uncertainty is represented as a corridor between and around them. [Toy simulation only shows the last one and uses the “slope and intercept toy simulation”]. With AIS data, this can be developed as an uncertainty corridor around nominal historical or predicted trajectories. Finally a comparison is presented for each cases- historical, predicted and historical+predicted>uncertainty.

Math Formulation

In the toy simulation, controlled line parameters changes were used instead of implementing full scale models like deep learning prediction models, Gaussian Process based uncertainty prediction, and uncertainty tube around nominal trajectories. Future prediction, trajectory uncertainty and combined historical plus predicted uncertainty were all represented by controlled changes in slope and intercept.

For straight lines $\mathbf{y}=\mathbf{m}\mathbf{x}+\mathbf{b}$, small changes were added to slope $\mathbf{m}$ and the intercept $\mathbf{b}$.

In the original paper, sensor placement is based on trajectory intensity estimated from historical data. So it first estimates where trajectories are most likely to occur and places sensors to reduce missed detection. The proposed extension keeps the same final objectives, just contributing more trajectory information.

Let us represent the trajectory intensity estimated from historical trajectories with $\lambda\mathbf{H}$, which was already discussed in the paper. If two more components are proposed, intensity estimated from predicted future trajectories as $\lambda\mathbf{F}$ and intensity estimated from uncertainty adjusted trajectories as $\lambda\mathbf{U}$, then the new trajectory intensity can be written as: $\lambda\mathbf{E} = \mathbf{w}_H\lambda\mathbf{H} + \mathbf{w}_F\lambda\mathbf{F} + \mathbf{w}_U\lambda\mathbf{U}$. Here, $\mathbf{w}_H$, $\mathbf{w}_F$, and $\mathbf{w}_U$ are weights that give priority consideration to each traffic set.

Then the optimization problem becomes:

$$S^* = \arg \min_{|S|=M} P_{\text{miss}}(S, \lambda_E)$$

So, choosing M sensor locations that minimize missed detection probability with the extended trajectory intensity.

MATLAB setup and simulation:

As mentioned in the technical part, since no AIS data were available, synthetic historical data were generated as straight lines with small changes with slopes and intercepts.

For all the cases, this same base workflow was used like- line representation> bijective transformation> representation space> sensor detection model> void probability idea> greedy search. Instead of implementing a full LGCLP and INLA, a simple gaussian smoothing was implemented to create an intensity map.

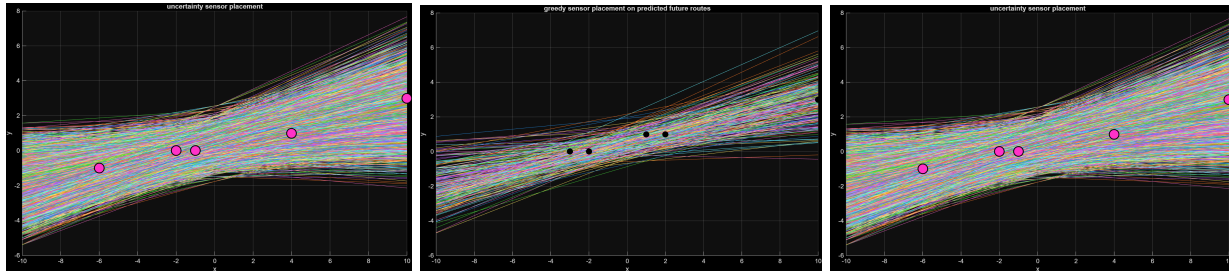
For prediction a simple toy regression was implemented using the same “slope-intercept toy” technique. Then, for the uncertainty aware case, historical and predicted routes were combined as a route set and again with the same “slope-intercept toy” technique, plausible deviations were created. The three cases were compared with selected sensor locations. For the entire

simulation, different parameter values were changed as input to make the results fit for better interpretation, because the data were synthetic.

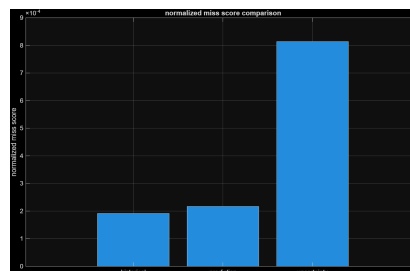
Results, code optimization and discussion:

Only the final comparison is shown here. All the other figures/results are presented in a separate file along with the code.

For toy simulation some parameters were adjusted to make the trajectory scenarios easier to compare.. Firstly, for prediction the same historical data were used that produced results that were narrow and hard to understand and compare. Then they were changed manually.



From left: 1. sensor placement with historical routes, 2. sensor placement with predicted future routes, 3. sensor placement for historical and predicted routes combined with uncertainty.



This second figure shows the normalized missed score comparison.

Since the data are synthetic, the comparison is not a performance evaluation but only shows the sensitivity of sensor placement due to different route sets and combinations. Also, the missed score is noticeably higher in the uncertainty case because the same number of sensors has to cover a wider possible route area. So, future work should consider prediction and wider uncertainty aware placement with more realistic context or condition based data.

References

1. H. Li, H. Jiao, and Z. Yang, "AIS data driven ship trajectory prediction modelling and analysis based on machine learning and deep learning methods," *Transportation Research Part E: Logistics and Transportation Review*, vol. 175, 103152, 2023. DOI: 10.1016/j.tre.2023.103152.
2. H. Rong, A. P. Teixeira, and C. Guedes Soares, "Ship trajectory uncertainty prediction based on a Gaussian Process model," *Ocean Engineering*, vol. 182, pp. 499 to 511, 2019. DOI: 10.1016/j.oceaneng.2019.04.024.